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Testing

Software Testing

Answer:

Software testing is the process of analyzing a software item to detect the differences between existing and expected conditions (bugs).

It ensures that the software works according to its requirements and helps identify defects before delivery.

Example:

Testing a login system by providing valid and invalid usernames and passwords to check whether it works correctly.

1. Black Box Testing

Answer:

Black box testing is a testing technique where the internal mechanism or code of a system is ***unknown to the tester.*** The tester focuses only on the inputs given to the system and the outputs produced.

Example:

Testing a login page by entering username and password without knowing the internal code.

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2. White Box Testing

Answer:

Testing a software while knowing the internal code and mechanisms, giving input and output to specifically test the internal parts of the code.

Example:

Unit test, Security test are examples of white box tests.

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3. Functional Testing

Answer:

Functional testing is a type of software testing where the system is tested against functional requirements specifications (FRS). It checks whether the software functions correctly by providing inputs and examining outputs.

Example:

Testing whether an e-commerce website correctly adds products to the shopping cart and completes checkout.

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4. Technical / Non-Functional Testing

Answer:

Technical testing checks the **non-functional aspects of software such as performance, usability, reliability, and security.** It focuses on how well the system performs rather than what functions it performs.

Example:

Testing how a website performs when thousands of users access it simultaneously.

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5. Unit Testing

Answer:

Unit testing is the testing of individual software units or groups of related units. It is generally performed by programmers to verify that each small part of the code works correctly.

Example:

Testing a calculator function separately to verify addition and subtraction operations.

Component Testing: Almost identical to unit testing, except at a slightly larger scale.

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6. Integration Testing

Answer:

Integration testing is performed when software components are combined and tested to evaluate their interaction. **It verifies that individual modules work correctly together after integration.**

Example:

Testing whether the login module properly connects with the user database.

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7. System Testing

Answer:

System testing is testing conducted **on a complete integrated system** to evaluate whether it satisfies specified requirements. It checks the entire software system including functional and non-functional requirements.

Example:

Testing a complete banking application after integrating all features.

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8. Load Testing

Answer:

Load testing is a performance testing method **that determines system performance under expected real-life load conditions.** It checks whether the system can handle the expected number of users and transactions.

Example:

Testing a university portal with 10,000 students accessing the system at the same time.

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9. Stress Testing

Answer:

Stress testing evaluates a system beyond its specified limits to determine its breaking point. **It checks how the system behaves under extreme load conditions.**

Example:

Increasing users on a website until the system becomes slow or fails.

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10. Security Testing

Answer:

Security testing identifies vulnerabilities in a system and ensures that data and resources are protected from unauthorized access. It checks the security strength of the software.

Example:

Testing whether hackers can access a user's account without proper authentication.

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11. Regression Testing

Answer:

Regression testing is **selective retesting of a system** after modifications to ensure that changes have not introduced new problems. ***It verifies that existing functionality still works correctly.***

Example:

After adding a new payment method, testing previous payment options again.

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12. Beta Testing

Answer:

Beta testing is when an organization provides a partial or complete version of software to potential users for real-world testing. Users report errors and feedback to improve the software.

Example:

A company releases a beta version of a mobile application to selected users before the final release.

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13. Acceptance Testing

Answer:

Acceptance testing is formal testing performed to determine ***whether a system satisfies customer acceptance criteria.*** It allows customers to decide whether the software is acceptable.

Example:

A customer tests a newly developed software system before approving final delivery.

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14. Static Testing

Answer:

Static testing ***refers to testing something that is not running.*** It involves reviewing and examining software artifacts such as code or documents.

Example:

Code review and software inspection.

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15. Dynamic Testing

Answer:

Dynamic testing involves executing the software, providing input values, and checking whether the output is as expected. ***It tests the behavior of a running system.***

Example:

Running a login test case with different usernames and passwords.